

Kunwar Abhuday Singh

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PROFILE

Software engineer specializing in AI/ML, DevOps, and Robotics. Built production AI agents, hardened enterprise CI/CD pipelines, deployed edge deep learning systems, and authored open-source ML tooling with measurable real-world impact.

EDUCATION

Manipal Institute of Technology

Bachelor of Technology, Computer Science and Engineering

Manipal, India

Jul. 2023 – Mar. 2027

EXPERIENCE

BookMyShow

Jun. 2026 – Jul. 2026

DevOps Engineer Intern

- Built the runtime for an internal **Agentic OS** that wires AI agents into company workflows, automating **15+** repetitive operational tasks and cutting manual turnaround time by **40%**.
- Hardened CI/CD and reliability tooling for internal platforms at ticketing scale, reducing deployment time by **30%** and freeing engineering hours across teams.

KLIV Group, IIT Kharagpur

Jan. 2025 – Present

Deep Learning Research Intern

Remote

- Optimized quantized CNN and Transformer models for edge deployment on Jetson Nano and Coral EdgeTPU via INT8 quantization and knowledge distillation, achieving **35% inference speed-up** at 90%+ accuracy and **40% model size reduction**.
- Benchmarked inference throughput across ONNX Runtime, PyTorch, and Intel Neural Compressor; characterized hardware-specific failure modes (ARM ISA incompatibility, OOM behavior) to guide deployment decisions.

Incubyte

Jun. 2025 – Jul. 2025

Software Engineering Intern

Remote

- Built and deployed **20+ enterprise AI agents** on **Azure** using the Agno framework across RESTful microservices for real corporate clients.
- Automated customer-support and business-analytics pipelines through agent orchestration, reducing manual workload by **60%** and increasing operational throughput by **40%**.

Team RoboManipal, MIT MAHE

Mar. 2024 – Present

Robotics Software Engineer

Manipal, India

- Developed real-time 6-DOF pose estimation with a custom ArUco pipeline in ROS and OpenCV at **20+ FPS**.
- Led ABU Robocon 2024 Nationals qualification, scoring **99/100** in Stage 1.

PROJECTS

Advance Conv Benchmarks: Open Source Python Library | [GitHub](#)

Python, PyTorch, TensorFlow, JAX

- Published PyPI library for FLOP/parameter counting and inference benchmarking across PyTorch, TensorFlow, and JAX; supports custom layer architectures. Accompanying research paper in peer review.

Ensemble Object Detection & 3D Pose Extraction | [GitHub](#)

Python, PyTorch, OpenCV, CUDA

- Built a distributed multi-GPU detection pipeline reaching **98%+ accuracy**; integrated depth-camera fusion for millimeter-precision 3D pose estimation with 40% latency reduction via parallel processing.

ROOT-ML Bridge: Physics & HPC Data Pipeline | [GitHub](#)

C++, Python, ROOT, RDataFrame, Parquet, XGBoost

- Built an end-to-end ML pipeline bridging CERN's ROOT framework with the Python ML ecosystem; implemented memory-efficient chunked processing via RDataFrame with automated provenance tracking and a full ROOT → Parquet → XGBoost round-trip.

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, Java

Frameworks: PyTorch, TensorFlow, HuggingFace, OpenCV, scikit-learn, MLflow, Docker, ROS

Cloud/Infra: Azure, Git, CI/CD, REST APIs, PostgreSQL, MongoDB, Parquet

Specialties: Computer Vision, NLP, Edge AI, INT8 Quantization, Knowledge Distillation, SLAM

RESEARCH & OPEN SOURCE

ML Benchmarking Paper: Comparing PyTorch, ONNX Runtime, and INC across x86-64, Raspberry Pi 5, and Jetson Nano (under review).

CERN ROOT Contributions: Fixed a data-corruption bug (ramtools PR #36); authored Issue #49 and draft PR #50.

ACHIEVEMENTS

RoboCup@Work Germany 2025: Top 8 internationally | **RoboCup@Work Brazil 2025:** Top 10 internationally | **Aether**

BytesBattle 2025: National Winners | **IISc RoboNautica 2025:** Runners-up